Nomenclature

ANCOVA Analysis of covariance

ANN approximate nearest neighbor

BGE BAAI General Embedding

DB Database

FAISS Facebook AI Similarity Search

FM Foundation Model

GTE General-purpose Text Embeddings

HF Hugging Face

HNSW Hierarchical Navigable Small World

IP Inner Product

KB Knowledge Base

LLM Large Language Model

MIP Maximum Inner Product

PQ Product Quantization

QA Question-Answering

RAG Retrieval Augmented Generation

Simi.Searches Similarity Search (Algorithm)

SustainabilityQA Sustainability Question-Answer Dataset

suswiki Sustainability Wikimedia

SVM Support Vector Machines

UAE Universal AnglE Embedding

0.1 Comparing LLM

In this experiment, we compare three different LLMs. They are Llama2 [1], Mistral [2], and GPT3.5 [3]. The goal of this experiment is to decide if LLM plays a role in answering the sustainability in the RAG model.

0.1.1 Prompt

We are unable to talk about LLM without talking about its input, the prompt. We are using a template prompt:

```
Use the following pieces of information to answer user's question.
If you don't know the answer, just say you don't know. Don't make up
   information yourself. Do not repeat yourself. Use the relevant
   information from the context to answer the question.
### Context: {contexts}
### Question: {query}
Only returns helpful answer below and nothing else.
```

In the template prompt above, we set up the prompt to use the context coming from the retrieval to help answer the question. The context is coming from the retriever, using the following parameters: KB - suswiki, text chunk - 200 characters, text overlay - 20 characters, embedding - BGE, vector store - FAISS, and distance index - euclidean. In this experiment, we are unable to use three top k documents and above due to the OpenAI token rate limitation.

With that said, we can approximate the length of our prompt using the Equation 1. Since we know that the maximum text chunk is 220 and the longest question in SustainabilityQA is 135 characters, we know that the maximum prompt as input is 907 characters or about 226 tokens [1]. In comparison, we limit the output token to 128 to keep the answer short.

$$\begin{aligned} l_{prompt} &= l_{prompttemplate} + k \cdot l_{textchunk} + l_{question} \\ &>= 332char + 2 \cdot 220char + 135char \\ &>= 907char \end{aligned} \tag{1}$$

0.1.2 Results and Discussion

Figure 1 shows that the difference between the three RAG models with different LLM is minimal. In terms of two RAGAS metrics of retrieval i.e. context recall and context precision, all three of the LLMs perform similarly, almost the same. This is expected since the same retrieval parameters are used in all three models. The slight variation is because RAGAS scoring is based on a generative AI model.

The answer relevancy, faithfulness, and answer correctness are metrics for the generative part of the RAG model. They measure the quality of the answer generated by the LLM. Answer relevancy measures the degree to which a response directly relates to the original question's

[1] meta-llama/Llama-2-13b-chat-hf
[2] mistralai/Mistral-7B-Instruct-v0.2
[3] gpt-3.5-turbo-0125

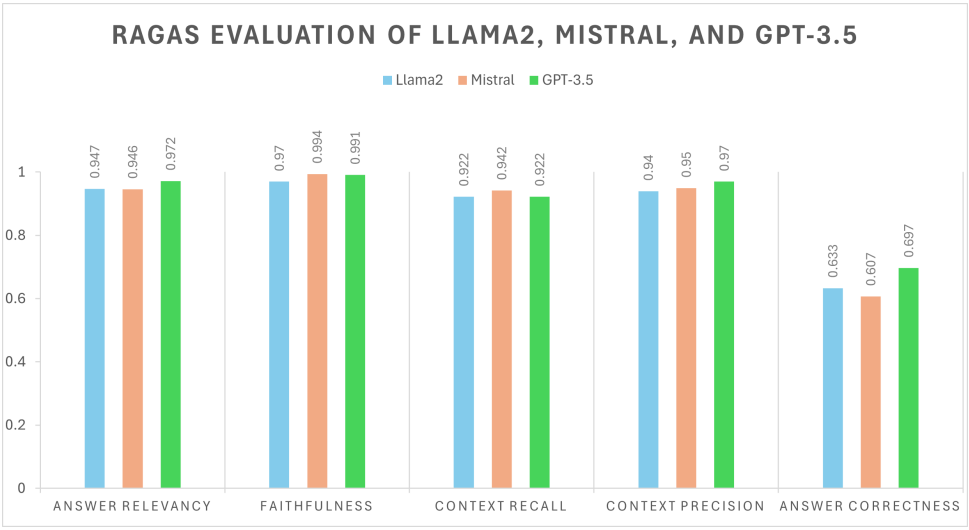


Figure 1: RAGAS evaluation of three different LLMs with the same retriever parameters

key points. Faithfulness measures the factual consistency of the generated answer against the given context from the retriever. In our experiment, both the answer relevancy and faithfulness scores of all three LLMs are similarly high, ranging from 0.946 to 0.994.

Answer correctness determines the accuracy of the generated answer compared to the ground truth. And since the ground truth is edited by humans, there are differences in the performance of these three LLMs. GPT3.5 has the highest RAGAS score with 0.697, followed by Llama2 and Mistral with 0.633 and 0.607 respectively. Although GPT3.5 has the highest score, it is still suboptimal, because according to the RAGAS author, a score less than 0.8 indicates a low degree of answer correctness.

The reason for low answer correctness is that the context has similar semantics, but not the same as the ground truth. Example: What is temporal autocorrelation? Context: Temporal autocorrelation, refers to the correlation between observations of a time series at different time points. Ground truth: Temporal autocorrelation is a principle that states humans value events in the near past or future more than those in the distant past or future. Several pages on suswiki talk about the same thing in different contexts.

In summary, changing the LLM between Llama2, Mistral, and GPT3.5 in the RAG model does not have a significant impact on the performance of the system. In terms of answer correctness, none of them reach the optimal value of 0.8. However, GPT3.5 has slight advantages in terms of answer correctness compared to Llama2 and Mistral.

0.2 Comparing All Retriever and Generation Parameters

In this section, we combine both different retriever parameters and different LLMs. The default parameters in our experiments are:

- Embedding: BGE
- Vector store: FAISS

- Index distance: Euclidean
- Top k document: 2

The purpose of default parameter choice is to have a baseline to compare the performance with the other parameters.

0.2.1 Results and Discussion

Overview Figure 2 shows the RAGAS metric scores of different combinations of retriever parameters, i.e. KB, embeddings, vector stores, and index distance, with different LLMs. The harmonic mean was used to compute the overall performance instead of the arithmetic mean in order to penalize more when a model has a low value. We created three new metrics, which are Retriever Harmonic Mean (Re.H.M), Generator Harmonic Mean (Ge.H.M), and RAGAS Harmonic Mean (H.Mean) to capture the quality of the retriever, generator, and everything respectively. Especially RAGAS harmonic mean can be used as a single-value representation to evaluate the score of the model.

At first glance, we can see that the model with suswiki KB has higher scores than a model with Wikipedia KB. This shows that KB has a huge impact on giving good results. Additionally, we can also infer that Wikipedia is statistically worse than suswiki in terms of sustainability.

In addition, GPT-3.5 consistently outperforms Llama2 and Mistral among the three LLMs. This is shown by the high score of GPT-3.5 compared to the similar scores of Llama2 and Mistral. This can be seen in the top right of Figure 3, where all of the points are GPT-3.5.

Speaking of Figure 3, we want to make it clear that the difference in scores between the different KBs (suswiki vs Wikipedia) is so large, that they create different clusters of data points in the top right and bottom left of the graph. The suswiki data points populate the upper right part of the graph, showing their absolute superiority over the Wikipedia KB. We can be sure that if we notice the same pattern in the other diagrams, the higher score consistently represents suswiki and vice versa for Wikipedia.

In Figure 4, we can observe different parameters being compared by the colors. However, this time the data is filtered by the default value, except for the data being compared. Again, the top right cluster is from suswiki KB and the bottom right is from Wikipedia. By splitting the data and comparing it using different parameters, we can see how different parameters play a role in the model quality. Although further analysis will be done separately in the subchapters below, Figure 4 can be used to check its value in a more granular way.

Knowledge Base Suswiki vs Wikipedia

As shown in Figure 3, the difference between suswiki and Wikipedia KB is very obvious. It is further proven by Figure 5, where the RAGAS harmonic means distributions are distant from each other. Suswiki leads with an average score of 0.86, while Wikipedia lags behind with 0.58.

KB	Emb.	V.S.	Index.	k	LLM	Co. R.	Co. P.	Ans. R.	Faith.	Ans. C	Re.H.M	Ge.H.M	H.Mean
Suswiki	**BGE**	FAISS	Eucl.	2	*Llama2*	0.922	0.940	0.947	0.970	0.633	0.931	0.818	0.860
Suswiki	**BGE**	FAISS	Eucl.	2	**Mistral**	0.942	0.950	0.946	0.994	0.607	0.946	0.809	0.858
Suswiki	**BGE**	FAISS	Eucl.	2	***GPT-3.5***	0.922	0.970	0.972	0.991	0.697	0.945	0.864	0.895
Suswiki	**BGE**	FAISS	Eucl.	1	*Llama2*	0.921	0.940	0.916	0.959	0.645	0.930	0.814	0.857
Suswiki	**BGE**	FAISS	Eucl.	1	**Mistral**	0.918	0.960	0.955	0.954	0.608	0.939	0.802	0.852
Suswiki	**BGE**	FAISS	Eucl.	1	***GPT-3.5***	0.935	0.960	0.979	0.972	0.718	0.947	0.871	0.900
Suswiki	**BGE**	FAISS	*I.P.*	2	*Llama2*	0.912	0.950	0.949	0.988	0.626	0.931	0.819	0.860
Suswiki	**BGE**	FAISS	*I.P.*	2	**Mistral**	0.922	0.970	0.947	0.994	0.612	0.945	0.812	0.860
Suswiki	**BGE**	FAISS	*I.P.*	2	***GPT-3.5***	0.922	0.970	0.974	0.981	0.700	0.945	0.863	0.894
Suswiki	**BGE**	FAISS	Cos.	2	*Llama2*	0.922	0.950	0.949	0.988	0.636	0.936	0.825	0.866
Suswiki	**BGE**	FAISS	Cos.	2	**Mistral**	0.922	0.950	0.947	0.994	0.604	0.936	0.807	0.854
Suswiki	**BGE**	FAISS	Cos.	2	***GPT-3.5***	0.922	0.950	0.970	0.977	0.699	0.936	0.861	0.889
Suswiki	**BGE**	Chroma	Eucl.	2	*Llama2*	0.912	0.930	0.931	0.967	0.636	0.921	0.815	0.854
Suswiki	**BGE**	Chroma	Eucl.	2	**Mistral**	0.912	0.950	0.927	0.988	0.606	0.931	0.802	0.849
Suswiki	**BGE**	Chroma	Eucl.	2	***GPT-3.5***	0.932	0.950	0.975	0.969	0.707	0.941	0.864	0.893
Suswiki	*GTE*	FAISS	Eucl.	2	*Llama2*	0.899	0.930	0.950	0.967	0.644	0.914	0.824	0.858
Suswiki	*GTE*	FAISS	Eucl.	2	**Mistral**	0.909	0.930	0.948	0.966	0.644	0.919	0.824	0.859
Suswiki	*GTE*	FAISS	Eucl.	2	***GPT-3.5***	0.905	0.940	0.975	0.969	0.740	0.922	0.880	0.896
Wikipe.	**BGE**	FAISS	Eucl.	2	*Llama2*	0.720	0.380	0.931	0.498	0.582	0.497	0.625	0.567
Wikipe.	**BGE**	FAISS	Eucl.	2	**Mistral**	0.740	0.370	0.858	0.640	0.579	0.493	0.673	0.588
Wikipe.	**BGE**	FAISS	Eucl.	2	***GPT-3.5***	0.700	0.380	0.827	0.758	0.601	0.493	0.716	0.606
Wikipe.	**BGE**	FAISS	Eucl.	1	*Llama2*	0.760	0.320	0.875	0.547	0.624	0.450	0.656	0.555
Wikipe.	**BGE**	FAISS	Eucl.	1	**Mistral**	0.770	0.320	0.860	0.606	0.596	0.452	0.668	0.561
Wikipe.	**BGE**	FAISS	Eucl.	1	***GPT-3.5***	0.770	0.320	0.730	0.743	0.578	0.452	0.675	0.564
Wikipe.	**BGE**	FAISS	*I.P.*	2	*Llama2*	0.720	0.380	0.931	0.549	0.594	0.497	0.655	0.581
Wikipe.	**BGE**	FAISS	*I.P.*	2	**Mistral**	0.710	0.390	0.846	0.668	0.575	0.503	0.679	0.596
Wikipe.	**BGE**	FAISS	*I.P.*	2	***GPT-3.5***	0.740	0.390	0.768	0.752	0.602	0.511	0.699	0.609
Wikipe.	**BGE**	FAISS	Cos.	2	*Llama2*	0.730	0.380	0.931	0.538	0.599	0.500	0.652	0.581
Wikipe.	**BGE**	FAISS	Cos.	2	**Mistral**	0.720	0.360	0.845	0.661	0.573	0.480	0.675	0.581
Wikipe.	**BGE**	FAISS	Cos.	2	***GPT-3.5***	0.740	0.390	0.809	0.642	0.613	0.511	0.678	0.599
Wikipe.	**BGE**	Chroma	Eucl.	2	*Llama2*	0.740	0.300	0.931	0.492	0.597	0.427	0.627	0.528
Wikipe.	**BGE**	Chroma	Eucl.	2	**Mistral**	0.730	0.350	0.834	0.594	0.581	0.473	0.652	0.566
Wikipe.	**BGE**	Chroma	Eucl.	2	***GPT-3.5***	0.740	0.320	0.750	0.723	0.577	0.447	0.674	0.560
Wikipe.	*GTE*	FAISS	Eucl.	2	*Llama2*	0.705	0.360	0.924	0.731	0.617	0.477	0.737	0.605
Wikipe.	*GTE*	FAISS	Eucl.	2	**Mistral**	0.705	0.360	0.869	0.660	0.600	0.477	0.692	0.586
Wikipe.	*GTE*	FAISS	Eucl.	2	***GPT-3.5***	0.735	0.370	0.828	0.699	0.607	0.492	0.700	0.599

Figure 2: Overview table of the different parameters of RAG model. The green color coding shows high scores, while red shows low. The blue background in the parameter marks non-default parameters. KB: Knowledge Base, Emb. : Embedding, V.S.: Vector Store, Index. : Index Distance, k: top k documents, Ans.R.: Answer Relevancy, Faith: Faithfulness, Co.R.: Context Recall, Co.P.: Context Precision, Ans.C.: Answer Correctness, Re.H.M: Retriever Harmonic Mean, Ge.H.M: Generator Harmonic Mean, H.Mean: Overall RAGAS Harmonic Mean.

The RAGAS harmonic mean is a single value that shows the quality of the model. It is calculated by taking the harmonic mean of all RAGAS parameters, i.e. context recall, context

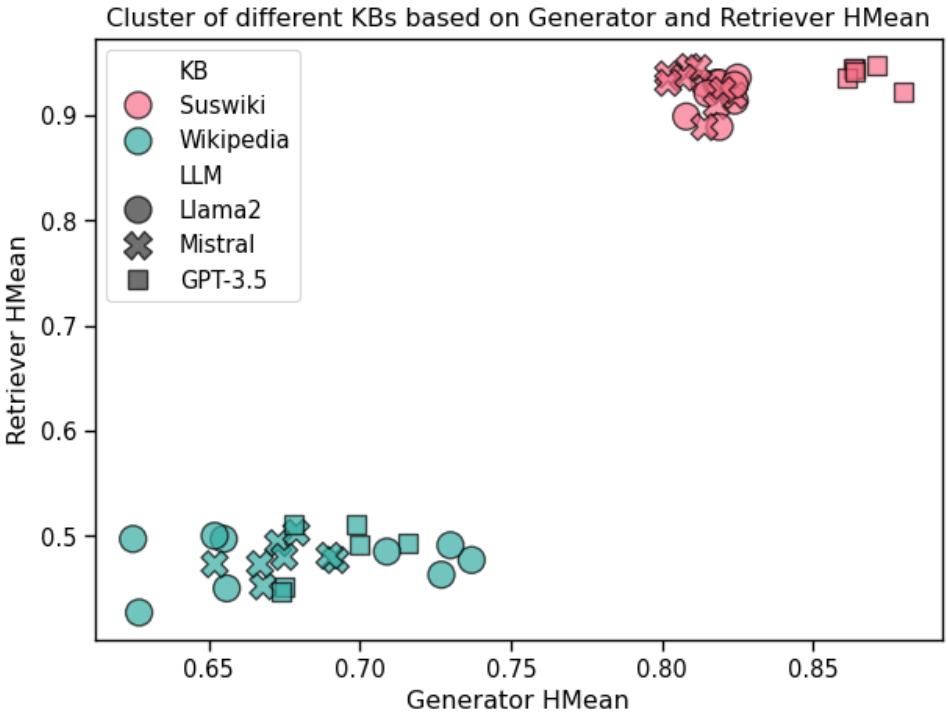


Figure 3: All points plotted in Generator H.Mean vs Retriever H.Mean. Suswiki's cluster is on the top right and Wikipedia's cluster is at the bottom left.

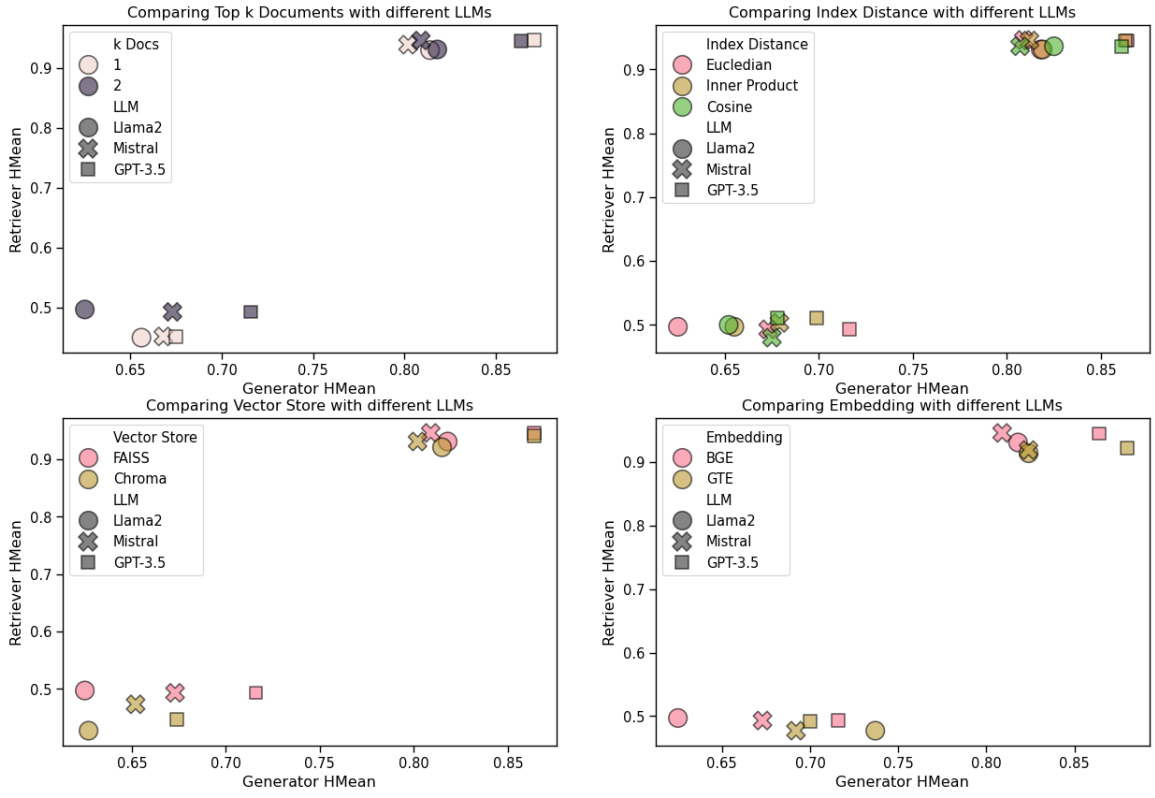


Figure 4: Scatter plot of different parameters compared to its default values.

precision, answer relevancy, faithfulness, and answer correctness.

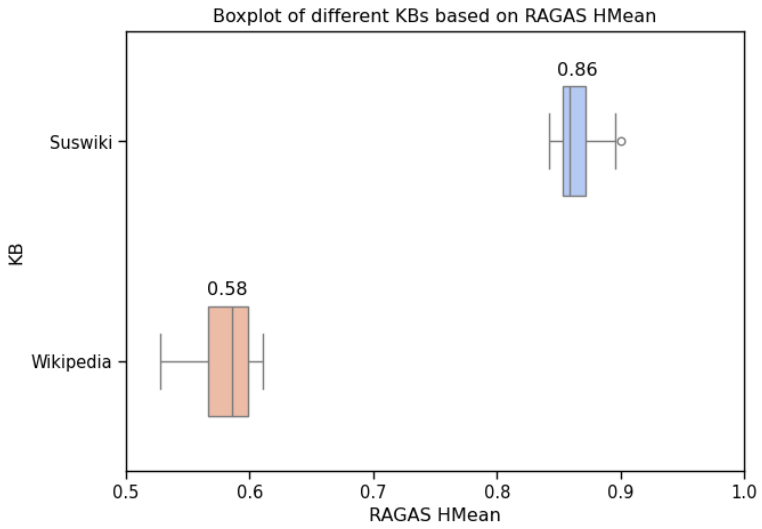


Figure 5: This box plot shows the distribution of RAGAS harmonic means between suswiki and Wikipedia. The numbers above the box plots show the average.

Embedding BGE [4] vs GTE [5]

Overall, GTE is slightly better than BGE with average harmonic scores of 0.734 and 0.729 respectively across the experiments with the default filter. There are 35 out of 50 questions where the retrieved contexts differ between BGE and GTE in the suswiki KB and 37 in the Wikipedia KB. Interestingly, the RAGAS score does not show a large discrepancy even though the embedding model is changed in Figure 6. This means that the contexts do not have to match the answer exactly, for the LLM to give a good answer. Close enough contexts added to the prompt will enable LLM to answer the question correctly since the LLM also has internal knowledge [2]. As proof, the largest discrepancy in the RAGAS score is shown by Llama2 with a difference of only 0.018.

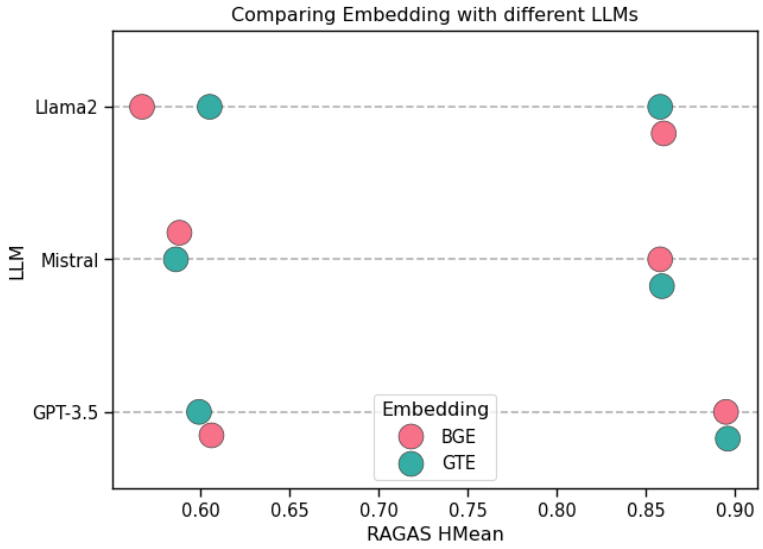


Figure 6: Overview of Different Embedding Performance in Different LLM.

[4]BAAI/bge-large-en-v1.5
[5]thenlper/gte-large

Vector Store FAISS[6] vs Chroma [7]

The overall average harmonic RAGAS score of FAISS is slightly better than Chroma with 0.729 and 0.709 respectively over the default parameters. This can be seen in Figure **??**. Also, in all three LLMs, FAISS gives better RAGAS scores to its LLM compared to Chroma. The difference in performance becomes more apparent in Wikipedia KB as compared to the suswiki KB. After comparing the context with FAISS and Chroma, we found that there are only two out of 50 different contexts, and therefore the changes in the RAGAS score are small.

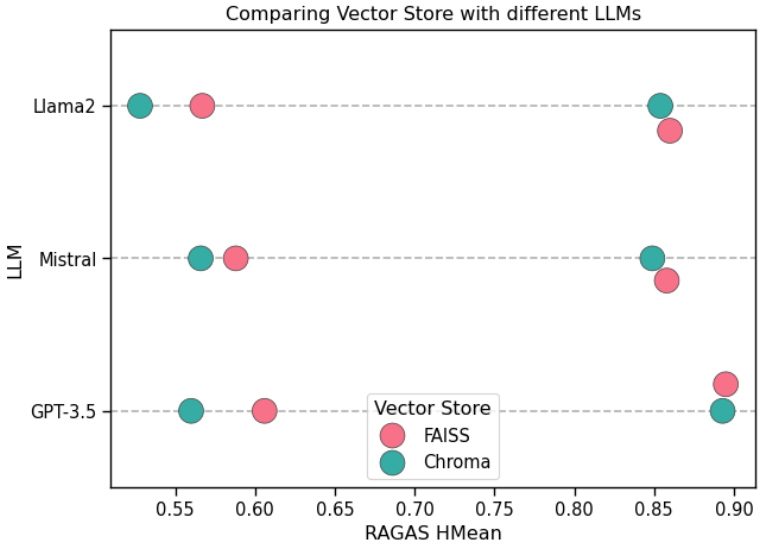


Figure 7: Overview of Different Vector Store Performance in Different LLM.

Index Distance Euclidean vs Cosine vs Inner Product

We can compare the overall performance of index distance by filtering the data with the default value k: two, embedding: BGE, and vector store: FAISS, resulting in the data shown in Figure 9. The average harmonic means of these three index distances are similar with the highest, *inner product*: 0.734, secondly, *euclidean*: 0.729, and the lowest, *cosine*: 0.728. As such, we think that the differences are negligible.

It is interesting to see in Figure 9 that LLMs have different preferences on different index distances. The combination of GPT-3.5 and cosine index gives the lowest score compared to Euclidean and inner product index. On the contrary, cosine gives Llama2 the highest score among other indexes. While Mistral prefers inner product compared to Euclidean and cosine.

This is rather strange since changing the index distance could mean changing the order of the retrieved contexts. After we checked, the top k documents, or in this case the top 2 retrieved contexts between different index distances are all the same. We assume that the small differences in scores are just caused by inaccurate RAGAS evaluation. However, further tests with different evaluators can be done to prove our hypothesis.

Top k Document One vs Two documents/contexts

[6]https://faiss.ai/index.html
[7]https://www.trychroma.com/

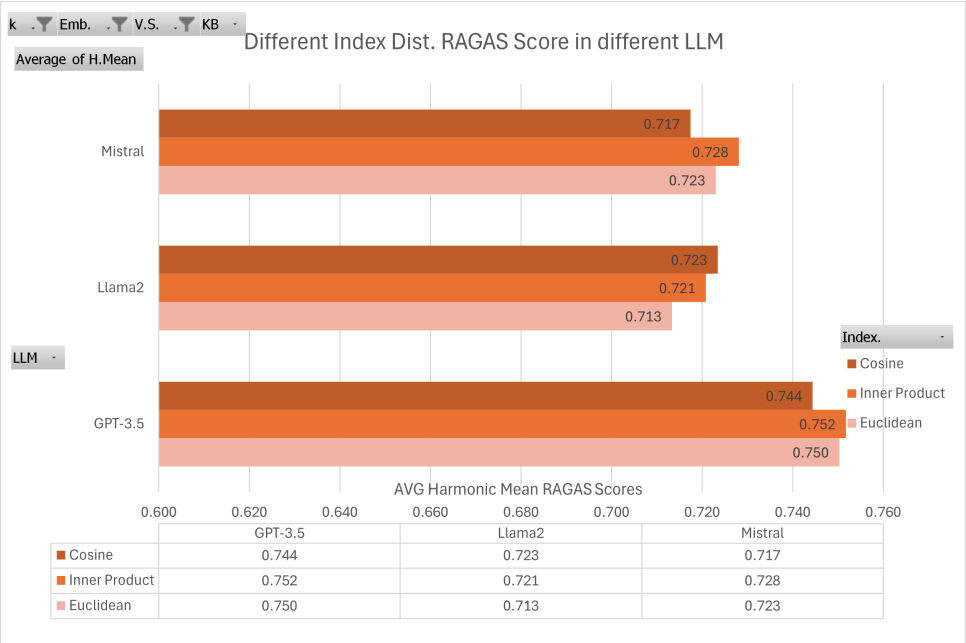


Figure 8: Bar Chart of RAGAS Harmonic Means by Different Index Distance Performance in Different LLM.

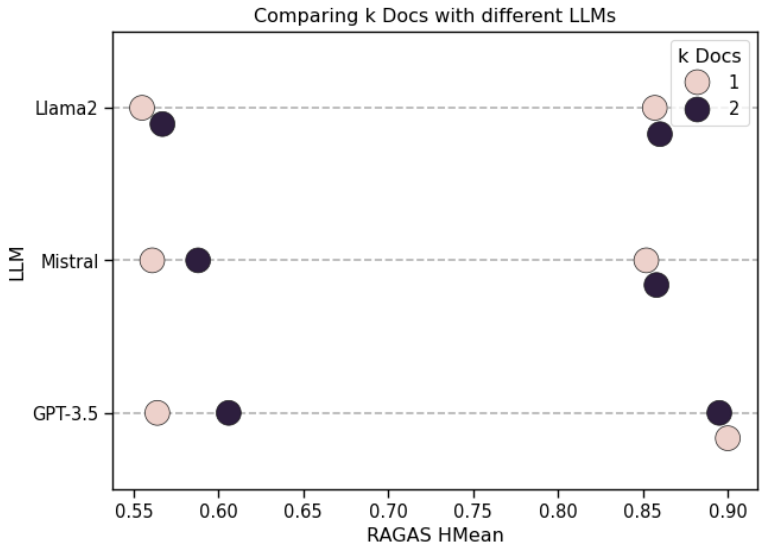


Figure 9: Overview of Different number of Top k Document in Different LLM.

When comparing *one* vs. *two* top k in the document, *two* top k generally give higher scores than *one*. However, in suswiki KB, the effect of retrieving more contexts does not affect the score that much. With GPT-3.5, retrieving *two* documents actually gives a lower score than retrieving only *one* document. Only when the information is general and more dense, like in Wikipedia, retrieving more context make an impact on the quality of the mode. So the number of top k documents here should be adjusted based on how large the KB is. A larger KB should have more number of k documents retrieved.

References

[1] *What are tokens and how to count them?* `https://openai.com/`, Accessed: 2023-04-12.

[2] F. Petroni, T. Rocktäschel, P. Lewis, *et al.*, *Language models as knowledge bases?* 2019. arXiv: `1909.01066 [cs.CL]`.